

Estimation of Muscle Cross-Sectional Area (ACSA) from Segmented Ultrasound Images of Lower Limb Muscles for Sports Performance Analysis

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Abstract—Sports science, rehabilitation, and musculoskeletal analysis should be concerned with accurate estimation of muscle anatomical cross-sectional area (ACSA) using ultrasound images. Manual segmentation is however time-consuming and also prone to observer variability. This paper presents a machine learning system to segment the Rectus Femoris muscle and predict ACSA based on ultrasound images automatically. It employs the preprocessing, multi-scale intensity, gradient, texture and spatial features to calculate the pixel-wise segmentation using a Random Forest classifier. A calibrated pixel to area factor is used to then translate the segmented muscle part into a physical cross-sectional area. It was demonstrated that the suggested method can be used to identify muscle areas with a high degree of confidence and estimate automated ACSA based on ultrasound imagery.

Index Terms—Muscle Segmentation, Ultrasound Imaging, Anatomical Cross-Sectional Area (ACSA), Random Forest, Feature Extraction, Texture Analysis, Image Processing, Medical Image Analysis, Sports Science, Musculoskeletal Assessment

I. INTRODUCTION

MUSCLE anatomical cross-sectional area (ACSA) is a significant biomarker in the field of sports science and rehabilitation to determine muscle strength, hypertrophy, fatigue, and risk of injury. Ultrasound is a non-invasive and readily available technique of imaging of muscles like Rectus Femoris. The measurement of ACSA of ultrasound image is however difficult because of speckle noise, low contrast boundaries and variation in image acquisition. Hand segmentation also takes time and is also prone to observer variation.

In this report, a traditional machine learning pipeline is created to segment muscle regions automatically and estimate ACSA using ultrasound images. The method involves hand-crafted intensity, texture, gradient, and spatial characteristics coupled with a Random Forest classifier to perform segmentation at pixel level. The proposed framework is based on the multi-scale image features with spatial priors, as it is intended to produce sound segmentation masks and facilitate automated cross-sectional area estimation of muscles.

II. METHODOLOGY

The dataset contained 338 valid ultrasound images of the Rectus Femoris muscle, and associated segmentation masks.

To partition the data, each image-mask pair was assigned to the training (70% set, $n = 236$), validation (15% set, $n = 50$), and test (15% set, $n = 52$).

Preprocessing was done by thresholding each image to extract the valid scanning region. Then, the image was cropped to the minimum bounding box and a median filter with a kernel size of five pixels was used to reduce the speckle noises. A CLAHE based adaptive histogram equalization algorithm was then applied with a clip level of 2.0 to increase contrast in the image. All preprocessing was performed at the original resolution.

The feature extraction step gave a 24-dimensional feature vector with every pixel belonging to six different groups. The local mean and standard deviation values were calculated with accordance to each scale (7×7 , 21×21 , 51×51 , and 71×71 pixels). Local brightness was evaluated by an intensity ratio measure of the local brightness in reference to the larger regional environment. The rank-order local entropy and variance estimates were calculated in a disk neighbourhood with a radius of 15 pixels. Additionally, Laplacian response and multi-scale directional gradients ($\sigma = 1$ and $\sigma = 3$) were added. Path based GLCM texture descriptors offered contrast, homogeneity, energy, and correlation scales of non-overlapping 32×32 pixel patches, and combined with normalized distances to scan boundary and centroid, allowed the representation of the spatial priors of muscle localization.

The sample construction of training involved a hard-negative mining approach of sampling 3,000 pixels of each training image. The composition consisted of 1,500 pmuscle pixels, 1,050 boundary negatives which were drawn on a band 5-50 pixels wide around the muscle margin, and 450 random background negatives outside the band width to preserve class balance and eliminate sampling of the background in the valid scan.

Random Forest classifier was utilized where each decision tree formed a hierarchy of binary splits on the values of the features, and the feature space was split until a prediction was reached. Minimization of the Gini impurity at every node was used to grow the trees. Raw output of the model, a binary pixel-based segmentation map that defined a pixel value as

1 for muscle and 0 for background, was morphologically closed with a 5-pixel radius, had holes filled in, connected components smaller than 500 pixels were removed, and the largest connected component was selected.

Dice coefficient and Intersection-over-Union (IoU) scores, which are calculated by comparing prediction segmentation masks and ground-truth masks at their native resolution, were used to evaluate the metrics.

$$Dice = \frac{2TP}{2TP + FP + FN} \quad (1)$$

$$IoU = \frac{TP}{TP + FP + FN} \quad (2)$$

The number of predicted pixels in the muscle was turned into a physical area measure by a calibration factor of cm^2 per pixel, and thus an ASCA value was found. Calibration factors were determined using the training images and the median value replaced the ground truth metadata in both validation and test sets to convert the number of pixels into ASCA estimates.

$$cm_{perpixel}^2 = \frac{Area_{gt}}{pixel_count_{gt}} \quad (3)$$

$$ACSA_{pred} = pixel_count_{pred} \times cm_{perpixel}^2 \quad (4)$$

III. RESULTS AND DISCUSSION

The proposed pipeline was tested on a dataset of 338 ultrasound images of lower limb muscles after removing 174 failed samples. The dataset was split at the image level into 236 training images (69.8%), 50 validation images (14.8%), and 52 test images (15.4%). From each training image, 3000 pixels were sampled (1500 muscle pixels, 1050 boundary negatives, and 450 random background pixels), resulting in approximately 708,000 labelled samples for model training.

For each sampled pixel, 24 features were extracted to capture intensity statistics, gradient information, texture descriptors, and spatial context. These included multi-scale mean and standard deviation values, entropy and variance, gradient magnitude, Laplacian responses, and GLCM texture features, along with spatial features such as distance from the muscle boundary and distance from the image center. These features were used to train a Random Forest classifier, which achieved an out-of-bag (OOB) score of 0.8738, indicating good generalization.

The Dice score and IoU on the validation set were 0.7475 and 0.6161 respectively. The model scored 0.6741 and 0.5454 in the test set with Dice and IoU respectively. Though these outcomes indicate a sensible segmentation performance, the Dice and IoU scores exhibit a certain fluctuation because it is

difficult to properly detect irregular boundary of muscles in ultrasound images and the speckle noise exists.

Visual results show that the predicted segmentation masks generally follow the overall shape of the muscle region and align reasonably well with the ground truth masks. Some errors occur in low-contrast regions or due to ultrasound speckle noise, which can lead to slightly irregular predicted boundaries.

After segmentation, ASCA was computed by converting the segmented pixel area into physical units using a calibration factor of 0.000189 cm^2 per pixel. Example predictions produced ASCA values ranging from approximately 23.8 to 24.7 cm^2 , demonstrating that the system can automatically estimate muscle cross-sectional area from ultrasound images.

In general, the findings indicate that the suggested feature-based Random Forest pipeline can be used to divide the muscles into regions and approximate ASCA with a decent accuracy, even though there are limited boundary errors due to ultrasound noise and inconsistency.

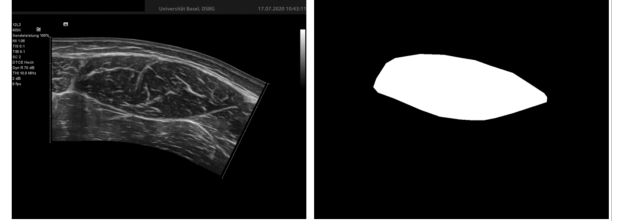


Fig. 1: Original Ultrasound (left) and Ground Truth Mask (right)

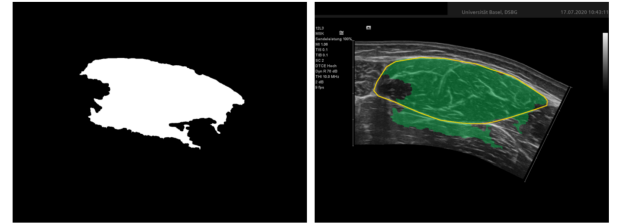


Fig. 2: Predicted Mask (left) and Overlay (right). ASCA value is 24.74 cm^2 , Dice score is 0.847, and IoU is 0.734.

IV. CONCLUSION

This report introduced a classical machine learning pipeline of automatic segmentation of the Rectus Femoris muscle in ultrasound images and determination of its ASCA value. The suggested system involved preprocessing, multi-scale feature analysis, and Random Forest classifier to conduct pixel-wise segmentation of the muscles. Each pixel was represented by a set of intensity, texture, gradient and spatial features and a hard-negative sampling approach enhanced the capacity of the model to differentiate muscle borders and tissues surrounding them.

On the validation set, the method had a Dice of 0.7475 and IoU of 0.6161 whereas on the test set, it had 0.6741 Dice and 0.5454 IoU. These findings suggest that the model can be used to predict the general muscle structure, but different images are segmented differently because of speckle noise, low contrast boundary, and the differences in the ultrasound scans. These restrictions notwithstanding, automatic estimation of ACSA values using the predicted masks was realized. Additional enhancements in the future can be to use more sophisticated texture descriptors or deep learning methods to get better boundary delineation.

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